SAMARTH

CUET PG 2027

Lecture – 09 | Computer Science

Unit 03 : Digital Fundamental

Topic : Boolean Algebra and Logic Gates: Part 2

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Recap *of previous lecture*

1. Boolean Algebra and Logic Gates: Part 1



Topics *to be covered*

1. Boolean Algebra and Logic Gates: Part 2



Topic : Boolean Algebra and Logic Gates: Part 2

#Q. The Boolean algebra is mostly based on

A. Boolean theorem

B. De Morgans theorem

C. Standard theorem

D. All of the above

Topic : Boolean Algebra and Logic Gates: Part 2

Ans. B

Solution:

De Morgans Law or De Morgans Theorem is the basics of the Boolean Algebra.

$$\textcircled{1}\ (A+B)' = A' \cdot B'$$

$$\textcircled{2}\ (A \cdot B)' = A' + B'$$

Topic : Boolean Algebra and Logic Gates: Part 2

#Q. Which among the following is not a Basic logical operator in Boolean Algebra laws?

A. AND
B. OR
C. NOT
D. NAND

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Ans. D

Solution:

NAND following is not a Basic logical operator in Boolean Algebra laws.

It is so because this is a combination of NOT and AND operator. There are only three basic logic operator named as OR, AND and NOT.

Topic : Boolean Algebra and Logic Gates: Part 2

#Q. Boolean algebra obeys-

A. commutative law only

B. distributive law only

C. associative law only

D. associative, distributive and commutative law

Topic : Boolean Algebra and Logic Gates: Part 2

Ans. D

Topic : Boolean Algebra and Logic Gates: Part 2

#Q. In Boolean algebra 1+1+1+1+.....800 times ones = _____.

A. 1

B. 0

C. 11

D. 800

1+1 = 1+1=1

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Ans. A

Solution:

In Boolean algebra '+' means ~~concatenation~~ OR. For any number of times '1' we will get 1 as the answer.

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X	X	X·X = X
0	0	0
1	1	1

#Q. In Boolean algebra, rule (X+Y)(X+Z) = ?

A. Y+XZ

B. X + YZ ✓

C. XY+Z

D. XZ + Y

$XX + XZ + XY + YZ$

$X + XZ + XY + YZ$

1	Z	1+Z
1	0	1
1	1	1

$X(1 + Z + Y) + YZ$

1

$X + YZ$

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Ans. B

Solution:

$(X+Y)(X+Z)$

$= X + XZ + XY + YZ$

$= X(1+Z+Y) + YZ$

$= X + YZ$

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#Q. Which of the following Boolean algebra rules is correct?

A. $A\bar{A} = 1$

B. $A + AB = A + B$

C. $A + \bar{A}B = A + B$

D. $A(A + B) = B$

(A) $A\bar{A} = 1$

A	$\bar{A}$	$A \cdot \bar{A}$	$A + \bar{A}$
0	1	0	1
1	0	0	1

(B) $A + AB$

$A(1 + B)$

$A \cdot 1$

$= A$

(C) $A + \bar{A}B$

$(A + \bar{A})(A + B)$

$1 \cdot (A + B)$

$A + B$

(D) $A(A + B)$

$= AA + AB$

$= A + AB$

$= A(1 + B)$

$= A$

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Ans. C

Solution:

Using transposition theorem:

$(A+A'B) = (A+A')\ (A+B)$

Since, $A+A' = 1$

$(A+A'B) = (A+B)$

Topic : Boolean Algebra and Logic Gates: Part 2

#Q. Which of the following Boolean algebra rules is correct?

A. $A.\overline{A} = 1$

B. $A+AB = A + B$

C. $A + \overline{A}B = A + B$

D. $A(A+B) = B$

Topic : Boolean Algebra and Logic Gates: Part 2

Ans. C

Solution:

1. $A.\overline{A} = 1$ incorrect as
2. A+AB = A + B is incorrect as A + AB = A
3. A + $\overline{A}$B = A + B is correct.
4. A(A+B) = B is incorrect as A(A+B) = A

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#Q. Consider a Boolean statement, x+x=x, what is this law called in Boolean algebra?

A. Idempotent law

B. Commutative law

C. Distributive law

D. Duality law

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Ans. A

Solution:

According to the idempotent law of Boolean algebra, $x+x=x$ and $x.x=x$, both are true.

Thus the correct option is A.

Topic : Boolean Algebra and Logic Gates: Part 2

#Q. Simplified the given Boolean algebra-

$(AB)'(A' + B)(B' + B)$

A. A'

B. A.

C. B

D. B'

$A'+B'$

B	B'	B+B'
0	1	1
1	0	1

$(A'+B')(A'+B)$

$= A'A' + A'B + A'B' + BB'$

A' $\quad$ 0

$A'(1 + B + B')$

A'

Topic : Boolean Algebra and Logic Gates: Part 2

Ans. A

Solution:

(AB)'(A'+B)(B'+B) = Given Original Expression.

(AB)'(A'+B)(1) = By using Complement Law, Identity Law.

(A'+B')(A'+B) = De Morgan's Law (AND with 1 was dropped because A*1 = A

A'A'+A'B+B'A'+B'B = Distributive Law.

A'+A'(B+B')+0 = Redundancy (on A's), Identities

A'+A'(1)

A'

So option A is correct answer.

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#Q. Which of the following is not valid Boolean algebra rule?

A. $X.X = X$

B. $(X + Y).X = X$

C. $X' + XY = Y$

D. $(X + Y).(X + Z) = X + YZ$

(A) $X \cdot X = X$

$\bar{X}$	X	X	$X \cdot X$	$X + \bar{X}$
1	0	0	0	1
0	1	1	1	1

(B) $(X+Y) \cdot X$

$XX + XY$

$X + XY$

$X(1+Y)$

$X \cdot 1 = X$

(C) $X' + XY$

$(X' + X)(X' + Y)$

$1 \cdot (X' + Y)$

$X' + Y$

(D) $(X+Y)(X+Z)$

$XX + XZ + XY + YZ$

$X + XZ + XY + YZ$

$X(1 + Z + Y) + YZ$

$X + YZ$

Topic : Boolean Algebra and Logic Gates: Part 2

Ans. C

Solution:

i) $X.X = X$

ii) $(X + Y).X$

$= X.X + X.Y$

$= X + X.Y$

$= X(1 + Y)$

$= X$

iii) $X' + XY$

$= (X' + X)(X' + Y)$

$= (1)(X' + Y)$

$= (X' + Y)$

iv) $(X + Y).(X + Z)$

$= X.X + X.Z + X.Y + Y.Z$

$= X(1 + Z + Y) + Y.Z$

$= X + Y.Z$

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#Q. Simplified the given Boolean algebra

(A+C)(AD+AD′) + AC + C

A. A+C

B. C+D

C. A′+C

D. A+D′

$A(D+D')$

$A \cdot 1$

$= A$

$(A+C) \cdot A$

$= AA + AC$

$= A + AC$

$= A(1+C)$

$= A \cdot 1$

$= A$

$A + AC + C$

$A(1+C) + C$

$A + C$

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Ans. A
Solution:

Expression	Rule(s) Used
(A+C)(AD+AD')+AC+C	Given Original Expressions.
(A+C)A(D+D')+AC+C	Distributive.
(A+C)A+AC+C	Compliment, Identity
AA+AC+AC+C	Distributive.
A+AC+C	Redundancy
A+C+C	Identity
A+C	Redundancy.

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#Q. Which of the following statements related to Boolean algebra is FALSE?

A. $X.(\bar{X} + 1) = X$

B. $X.(\bar{X} + \bar{X}Y) = 1$

C. $X + \bar{X} = 1$

D. $1 + X = 1$

(A) $X(\bar{X}+1)$
$= X\bar{X} + X$
$0 + X$
$= X$

(B) $X(\bar{X} + \bar{X}Y)$.
$= \underbrace{X\bar{X}}_{0} + \underbrace{X\bar{X}}_{0}Y$
$= 0$

(C) $X + \bar{X} = 1$

X	$\bar{X}$	$X+\bar{X}$	$X \cdot \bar{X}$
0	1	1	0
1	0	1	0

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Ans. B

Solution:

(A) $X.(\overline{X}+1) = (X.\overline{X}+X) = X$ Given Statement is correct

(B) $X.(\overline{X}+\overline{X}Y) = (X\overline{X}+X\overline{X}Y) = 0$ Given Statement is incorrect

(C) $X + \overline{X} = 1$, Given Statement is correct

(D) $1 + X = 1$, Given Statement is correct

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#Q. The complete set of only those Logic Gates designated as universal Gates is

A. NOT, OR and AND gates

B. XNOR, NOR and NAND Gates

C. NOR and NAND Gates

D. XOR, NOR and NAND Gates

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Ans. C

Solution:

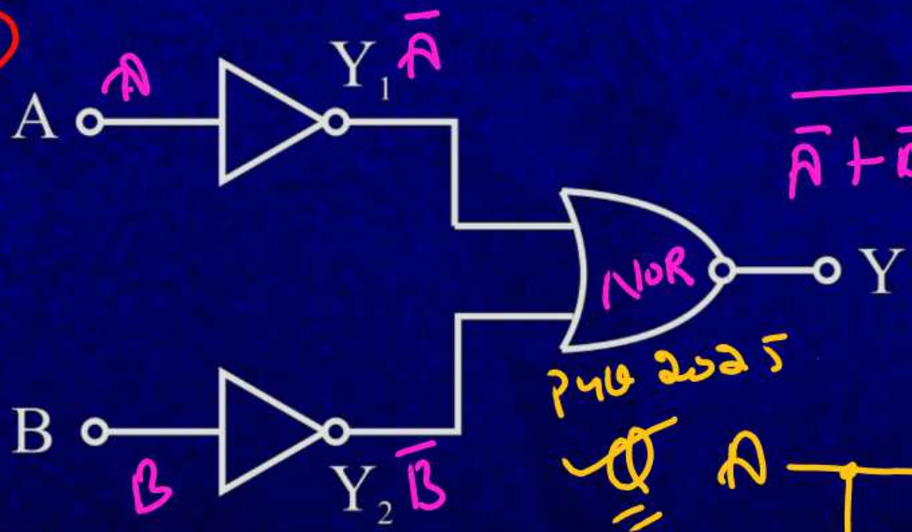
As per the logic gate combination possibility any logic gate can be made using NOR and NAND gate.

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#Q. Which logic gate is represented by the following combination of logic gates?

$XOR = A \oplus B = A'B + AB'$

$XNOR = A \odot B = \bar{A}\bar{B} + AB$



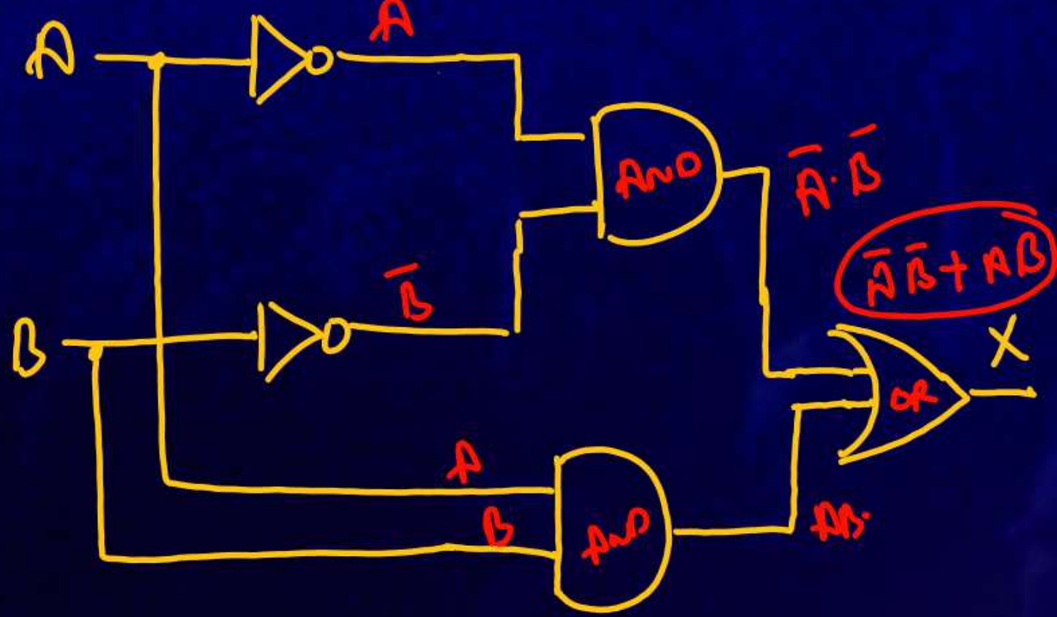
$\overline{\bar{A} + \bar{B}} = \bar{\bar{A}} \cdot \bar{\bar{B}} = AB$

Put 2 and 3

A. OR
B. NAND
C. AND
D. NOR

The full circuit generates same O/P as :—

(1) XOR (2) XNOR
(3) NAND (4) NOR



Ans. C

Solution:

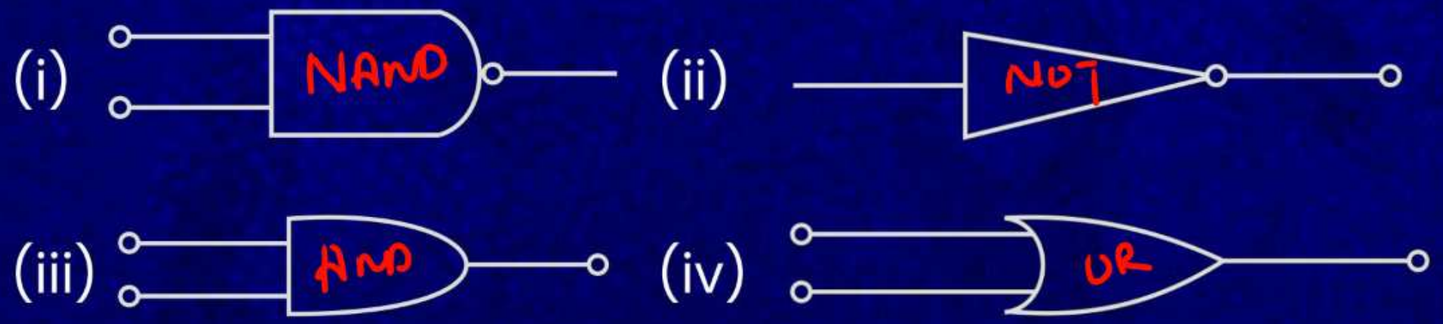
The truth table for the given circuit is

A	B	Y_1	Y_2	Y
1	1	0	0	1
1	0	0	1	0
0	1	1	0	0
0	0	1	1	0

From the above truth table output of the combination high when both inputs are high. Hence, the combination represents AND gate.

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#Q. The symbolic representation of four logic gates



The logic symbols for OR, NOT and NAND gates are respectively

A. (iii), (iv), (ii)

B. (iv), (i), (iii)

C. (iv), (ii), (i)

D. (i), (iii), (iv)

Ans. C

Solution:

The option (c) gives the correct sequence of the OR, NOT (inverting logic gate) and the NAND logic gates.

Universal gates

(1) NAND

(2) NOR

Using these gates we can draw any gate.

Ex (1) NOT gate



(2) AND



Ex AND.

$$A \cdot B = B \cdot A$$

2 Mins Summary

LEARN

Implementation of Gates using NAND & NOR

	NAND	NOR
NOT	1	1
AND	2	3
OR	3	2
XOR	4	5
XNOR	5	4

LEARN

Properties of Gates

	Idempotent / Closure	Commutative	Associative
NOT $\sim$	X	NA	NA
AND $\cdot$	✓	✓	✓
OR $+$	✓	✓	✓
NAND $\uparrow$	X	✓	X
NOR $\downarrow$	X	✓	X
XOR $\oplus$	X	✓	✓
XNOR $\odot$	X	✓	✓

Homework from SAMARTH CUET PG 2027 Module

Summary :-

Q Practice



THANK YOU